

Citadel

The Knowledge Compilation Layer for Enterprise AI

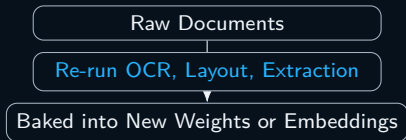
Masum Billal

August 3, 2026

Every AI System Re-Pays the Same Hidden Cost

Every AI system couples enterprise data to one model's representation — its weights or its embedding space. Both approaches skip the same intermediary step, and both pay for it the same way on every model swap.

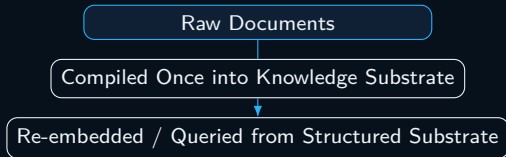
Without a Knowledge Layer



Full reingestion cost, paid again on every model swap.

Models iterate weekly. Applications pivot monthly.

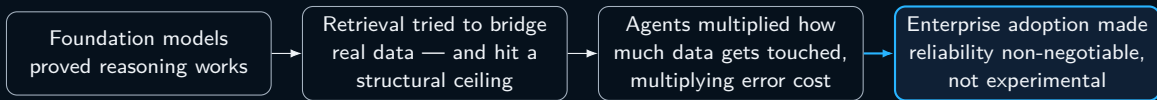
With Citadel



Extraction cost paid once. Model swaps touch only the substrate.

Infrastructure persists for decades.

Why Now: The Bottleneck Shifted from Models to Data Representation



The missing layer was survivable when AI was experimental. It stopped being optional the moment reliability became the product.

The Foundational Layer Every AI Infrastructure Requires

AI systems reasoning over real documents face the same dependency issue. An application built on a compiled knowledge substrate holds a compounding execution advantage.

Not a Feature. A Survival Dependency.

- ▶ Downstream uses lossy extraction.
- ▶ **The gap compounds:** A Citadel-backed system re-embeds in minutes on any schema or model change; legacy systems re-extract from scratch.

Who Relies On This

- ▶ **Enterprise AI Teams:** Building mission-critical agents over unstructured document stores.
- ▶ **Vertical AI Vendors:** Needing zero-hallucination data pipelines for legal, finance, and regulatory compliance.

Models compete on reasoning. Citadel changes what they are able to reason over.

Why Existing Infrastructure Cannot Solve This

Take excel as an example.

Frontier Model Vendors

(OpenAI, Google, Anthropic, Meta, etc.)

- ▶ **Incentive:** Token consumption & compute scale.
- ▶ **Abstraction:** Text token streams & context windows.
- ▶ **Blindspot:** Force structured tables into uncompiled token sequences.

Notable attempts

- ▶ TaBERT by Facebook AI
- ▶ TAPAS, TARL by Google
- ▶ TUTA, TaPEX by Microsoft
- ▶ LLMs and embedding

Mistake: force excels/tables into transformers

Failures and Reason

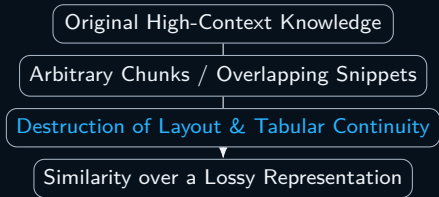
- ▶ Quadratic attention
- ▶ Tables are not language
- ▶ Retrieval mismatch
- ▶ Context explosion
- ▶ Different attack methods, same failure

Citadel owns the missing abstraction between raw storage and generative reasoning avoiding the root failure cause altogether.

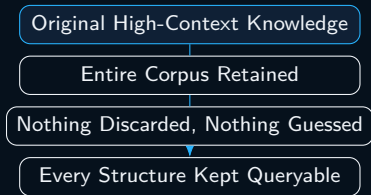
Retrieval Cannot Recover What Structurally Disappeared

If a representation discards structure, nothing downstream recovers it. Every answer must trace back to exactly where it came from — nothing flattened, nothing untraceable.

The Retrieval Failure Mode



The Citadel Paradigm



Citadel gracefully handles large document corpora, compiling everything before reasoning.

Proof of Failure: Unstructured Context Loss (ChatGPT)

ChatGPT

so how many tables did you say the excel file had and how many rows in each one?

From this workbook, I identified **3 logical tables** on the single worksheet.

Table	Approximate Rows	Number of Rows	Description
Table 1	3-18	16 rows	Black-Scholes option pricing model (lookup table, inputs, outputs, calculations)
Table 2	22-24	3 rows	Dataset metadata (World Development Indicators source information)
Table 3	25-62190	62,166 rows	World Development Indicators dataset

Total

- Worksheets: 1
- Logical tables: 3
- Total rows across all tables: **62,185 rows**

Notes

The third table is by far the largest:

- Rows:** 62,166
- Columns:** roughly 70 (Country Name, Country Code, Indicator Name, Indicator Code, followed by yearly columns from 1960 through 2025)

The first table is the trickiest because it's **not a clear tabular dataset**. It's really a financial worksheet composed of several related blocks (normal distribution lookup, input parameters, calculated values, call/put spreads, etc.). And it's somewhat messy with additional 1-5 rows of tables, but it's somewhat messy.

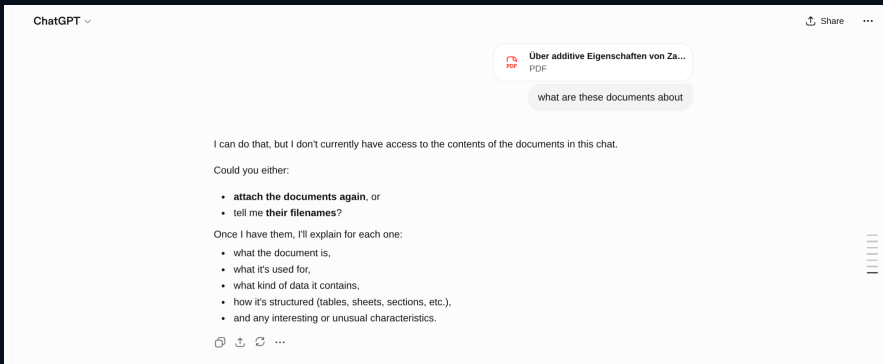
Failure Analysis: ChatGPT

Test: Tabular aggregation query across uploaded files (1 Excel + 809-page Number Theory book).

Failure: System hallucinated **60,000+ non-existent rows** (Ground Truth: 5,733 rows).

Root Cause: Standard context windowing and ungrounded text inference lose structural continuity across document boundaries.

Proof of Failure: Corpus Description Collapse (ChatGPT)



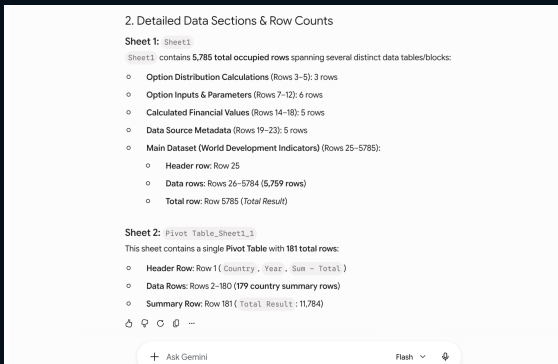
Failure Analysis: ChatGPT Overview Query

Test: Isolated Excel file uploaded first, followed by an 809-page Number Theory textbook; prompted with *"What are these documents about?"*.

Failure: System completely failed to respond accurately or describe the uploaded files.

Root Cause: Ingesting massive uncompiled PDF volumes alongside tabular data overloads context processing, causing parsing collapse.

Proof of Failure: Cross-File Schema Join Failure (Gemini)



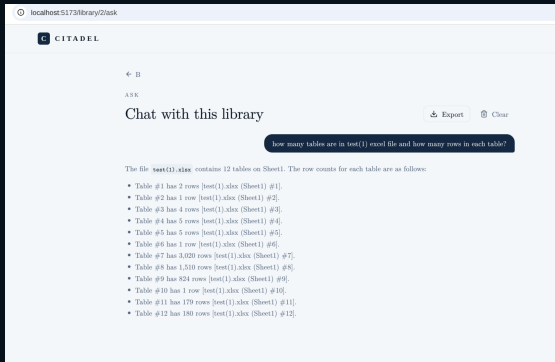
Failure Analysis: Gemini

Test: Stepwise ingestion (Excel file uploaded isolated first, followed by 4 PDFs).

Failure: Successfully described files, but returned **5,785 rows** on the query (Ground Truth: 5,733).

Root Cause: While capable of file-level summaries, semantic retrieval over uncompiled text fails exact cross-file table joins.

Proof of Execution: Citadel Ground Truth Execution



Citadel Execution Mode

Test: 29 multimodal document corpus ingested simultaneously (including the 809-page Number Theory PDF).

Result: 5,733 / 5,733 rows verified. Bounded factual execution.

Mechanism: Compiled into a canonical substrate → executed structural query directly over preserved schema across all 29 files. Zero generative

hallucination: factual outputs originate directly from the substrate.

Representation Beats Parameter Scale

Metric / System	ChatGPT (Frontier)	Gemini (Frontier)	Citadel (Local Node)
Compute Context	Enterprise Cluster	Enterprise Cluster	Single 5090 Laptop
Model Scale	~1T+ Parameters	Trillion Scale	9B Local Model
Tables	3 (isolated)	6 (isolated)	12 (All files)
Rows	60k+	5,785	5,733/5,733
Row Status	Fabricated	Structurally Drifted	Same as ground truth

Infrastructure Takeaway

Takeaway: Intelligence scale cannot recover structure destroyed during ingestion. By compiling documents into a deterministic substrate, Citadel enables a local 9B model running on consumer hardware to outperform trillion-parameter frontier systems on complex enterprise data.

Value Capture: From AI Assistance to Delegated Execution

Enterprises do not pay top dollar for text summaries. They pay to automate high-liability workflows where errors carry massive financial or regulatory consequences.

Current Bottleneck: Assistance Only

- ▶ Models produce probabilistic drafts.
- ▶ Humans must manually re-verify every row and source clause.
- ▶ **Result:** High human labor cost remains intact due to liability risk.

The Citadel Unlock: Delegated Workflows

- ▶ **M&A Due Diligence:** Cross-contract risk auditing with full row provenance.
- ▶ **Financial Audits:** Zero-loss reconciliation across hundreds of filings.
- ▶ **Regulatory Compliance:** Verifiable, zero-hallucination compliance queries.

Citadel converts AI from a probabilistic assistant into an accountable enterprise worker.

GTM: Land in High-Liability Workflows, Expand to Platform



Why Legal/M&A First?

- ▶ Highest density of complex tables, nested PDFs, and formulas.
- ▶ Errors carry direct financial/regulatory liability.
- ▶ High initial willingness to pay (\$250k+ ACVs).

The Platform Flywheel

- ▶ Substrate deployment creates lock-in across enterprise document stores.
- ▶ Every new internal enterprise AI application reuses the existing Citadel substrate.

Business Model: Enterprise Pricing & Unit Economics

Phase 1: Enterprise Licensing (Direct Value)

- ▶ **Base Platform Fee:** \$150,000 / year
(Includes core ingestion engine & 1M pages compiled).
- ▶ **Volume Expansion:** \$0.05 - \$0.15 per additional compiled page retained.
- ▶ **Target ACV:** \$250,000 - \$500,000 per enterprise node.

Phase 2: Platform Metering (API Scale)

- ▶ **Substrate Ingestion:** Billed on raw volume ingested and retained.
- ▶ **Query Execution:** Billed per deterministic query/retrieval call executed by downstream LLM agents.

Path to \$50M ARR

$$100 \text{ Enterprise Customers} \times \$500\text{k Average ACV} = \$50\text{M ARR}$$

The Ask: Capitalizing Enterprise Deployment

Raising **\$3,000,000** for **10%** equity (\$30M post-money valuation) — priced for a working system, not a concept.

What This Capital Funds

- ▶ Full-time founder engineering focus.
- ▶ First hires: 2x Senior Systems Engineers, 1x Enterprise GTM Architect.
- ▶ High-availability cloud infrastructure for enterprise pilots.
- ▶ Security, SOC2 compliance, and IP protection for enterprise deployments.

Milestones (18–24 Months)

- ▶ Technical risk fully retired — working core converts into enterprise deployments.
- ▶ Secure 510 paying enterprise design partners in legal/financial audits.
- ▶ Target: \$1M\$2M ARR baseline entering Series A.

Team & Scaling

Founder-Market Fit: First-Principles Systems Engineering

Masum Billal — Founder & Primary Architect

- ▶ **Identified Upstream Failure:** Recognized that standard RAG architectures suffer from structural information loss prior to model inference.
- ▶ **End-to-End Execution:** Architected the entire Citadel stack: multi-file schema parser, canonical knowledge substrate, and deterministic SQL engine.
- ▶ **Proven Leverage:** Built a working substrate that out-executes trillion-parameter cloud architectures on a single local GPU.

Phase 0: Systems Architecture (Completed)

Solo-built ingestion engine, multi-file table parser, canonical substrate, and deterministic query pipeline. Every benchmark in this deck runs on working code.

Phase 1: Enterprise Scaling (This Round)

Transitioning to enterprise deployments: expanding cloud reliability, onboarding initial design partners, and industrializing customer data integrations.

Built the missing abstraction layer from first principles.